Aryan Naik

aryannaik1205@gmail.com | github.com/RobotsAndAryan | UCL MEng Robotics and Artificial Intelligence (2025-2029)
UK national - eligible for Security Clearance

EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2025-2029

First year (applying for Summer 2026 internship). Modules include embedded systems, electronics, mechanical design, and control systems.

PROJECTS

Aerospace Avionics Firmware - Student Rocketry Telemetry Payload

- Reduced firmware complexity and improved execution determinism across a dual-MCU embedded architecture by implementing an interrupt-driven finite state machine (FSM) for autonomous parachute deployment, eliminating polling overhead across all flight-critical state transitions.
- Achieved sensor data acquisition at the IMU's full output rate by integrating IMU and barometric sensors over I2C, enabling continuous attitude estimation, altitude sensing, and acceleration logging throughout the flight envelope.
- Maintained onboard data logging and wireless telemetry transmission within a 3.3V supply rail by designing the power budget allocation across all active peripherals, ensuring no brownout events during flight.
- Supported post-flight verification by implementing UART/SPI serial communication between MCUs and flash storage, producing a complete recoverable dataset for every test flight conducted.

Full-System Robotics Hardware Integration - Competition Build

- Delivered a fully functional robotic platform within a fixed 48-hour build window by integrating microcontrollers, motor drivers, ultrasonic and IR sensors, and power electronics on a breadboard from schematic to physical assembly.
- Resolved all pre-competition hardware faults within the build window by applying systematic fault-finding and signal integrity checks across the sensor-actuator feedback loop, achieving stable closed-loop control prior to competition entry.
- Produced structural enclosures and mounting geometry to a dimensional tolerance of +/-0.5mm by completing mechanical CAD design in Autodesk Fusion 360 and fabricating parts via 3D printing.
- Reduced integration risk by applying component selection, voltage regulation, and signal conditioning discipline across the full electronics stack, verifying each subsystem independently before system-level acceptance testing.

SKILLS

Electronic and Embedded: C++, embedded firmware, real-time execution, interrupt-driven architecture, FSM design, I2C, SPI, UART, sensor fusion, IMU integration, signal conditioning, voltage regulation, motor control (H-bridge, servo), microcontroller integration

Mechanical: CAD (Autodesk Fusion 360), structural component design, enclosure design, tolerancing, 3D printing, rapid prototyping, component selection

Systems Engineering: hardware-software co-design, systems integration, subsystem verification and acceptance testing, fault-finding, closed-loop feedback control, power budgeting

Aerospace and Defence: avionics firmware, flight computer design, guidance navigation and control (GNC), autonomous recovery systems, safety-critical embedded software, rocketry telemetry

Generalist: breadboard electronics, mechatronics, schematic-to-assembly build, cross-discipline engineering, competition engineering under time and resource constraints

ADDITIONAL

- Actively developing skills in manufacturing processes and systems engineering methodology (requirements definition, verification and validation) through coursework and project work.
- Available for full summer internship period | Rochester location confirmed suitable.

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Hiring Team BAE Systems Rochester

Dear Hiring Team,

BAE Systems occupies a unique position at the intersection of aerospace engineering, defence electronics, and safety-critical systems integration. Your involvement in major multinational programmes spanning air, land, sea, and space domains, combined with your reputation for engineering excellence in advanced avionics and embedded defence electronics, makes this placement particularly compelling to me. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am eager to contribute to and learn from an organisation operating at this level of technical complexity.

My most directly relevant experience is the development of aerospace avionics firmware for a student rocketry telemetry payload. I designed and implemented C++ firmware on a dual-MCU embedded architecture featuring interrupt-driven real-time execution, I2C sensor integration for IMU data acquisition and altitude sensing, and a finite state machine governing autonomous parachute deployment under safety-critical flight conditions. I also implemented wireless telemetry transmission and onboard flash data logging. This project required me to balance hardware-software co-design, power budgeting, and signal integrity across a complete flight computer stack, producing a functional avionics system validated against real launch conditions.

I would bring hands-on embedded systems experience, a rigorous approach to safety-critical design, and the ability to work across mechanical, electronic, and software domains simultaneously. At BAE Systems, I am keen to develop my understanding of how defence-grade engineering standards and systems engineering processes are applied at scale, particularly within avionics and guidance, navigation, and control disciplines. I am confident this placement would meaningfully accelerate my development as an engineer.

Yours sincerely, Aryan Naik